

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : Civil Engineering Semester : VI Subject Code & Name: BTCVE605A Waste Water Treatments Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	Draw the flow diagram for preliminary treatment of sewage .State the type of impurities removed in each unit	CO1	6
B)	Explain Concept of sewage, sullage and storm water and Necessity of treatment of waste water.	CO1	6
C)	Explain in detail activated sludge process.	CO1	6
Q.2	Solve Any Two of the following.		
A)	Draw neat diagram & describe the self-purification zones of stream.	CO2	6
B)	What is D.O. sag .Explain process of natural purification by D.O. sag curve.	CO2	6
C)	Describe treatment process of oxidation ditch.	CO2	6
Q. 3	Solve Any Two of the following.		
A)	Explain in detail Physical, Chemical and Biological properties of Industrial Wastes water.	CO3	6
B)	Differentiate between sewage and Industrial wastewater.	CO3	6
C)	Enlist methods used for neutralization of acidic and alkaline wastes. Explain any one method of each in detail.	CO3	6
Q.4	Solve Any Two of the following.		
A)	Differentiate between Physical adsorption and Chemical Adsorption.	CO4	6
B)	Explain Nitrification and Denitrification.	CO4	6
C)	Explain Phosphorus removal from wastewater.	CO4	6
Q. 5	Solve Any Two of the following.		
A)	Explain Prevention of Communicable Diseases	CO5	6

B)	Explain Mosquito life cycle and explain Mosquito Control measures	CO5	6
C)	Explain in detail low cost excreta disposal systems	CO6	6
	*** End ***		

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